

WTW 158 - FIRST SEMESTER 2013

Sample problems and solutions - Solving inequalities .

There are more than one way to solve an inequality of the form $\frac{f(x)}{g(x)} < 0$ or $\frac{f(x)}{g(x)} > 0$.

You can use a table or you can use graphs. We will expect you to write down an algebraic solution . In these solutions I show you what I expect from you in a test.

- **Solving inequalities of the form** $\frac{f(x)}{g(x)} < 0$ or $\frac{f(x)}{g(x)} > 0$ or...

Example 1

Solve $\frac{x-2}{2x-3} \leq 1$ and write your answer in interval notation

Solution

$$\frac{x-2}{2x-3} \leq 1$$

$$\Rightarrow \frac{x-2}{2x-3} - 1 \leq 0 \text{ (I need the zero on the left!)}$$

$$\Rightarrow \frac{x-2-2x+3}{2x-3} \leq 0 \text{ (Simplify!)}$$

$$\Rightarrow \frac{-x+1}{2x-3} \leq 0$$

$$\Rightarrow -x+1 \leq 0 \text{ and } 2x-3 > 0 \text{ or } -x+1 \geq 0 \text{ and } 2x-3 < 0$$

$$\Rightarrow x \geq 1 \text{ and } x > \frac{3}{2} \text{ or } x \leq 1 \text{ and } x < \frac{3}{2}$$

$$\Rightarrow x > \frac{3}{2} \text{ or } x \leq 1$$

$$\Rightarrow x \in (-\infty, 1] \cup (\frac{3}{2}, \infty)$$

- **Solving quadratic inequalities**

You must use a graph to solve a quadratic inequality/

Solve and write your answer in interval-notation.

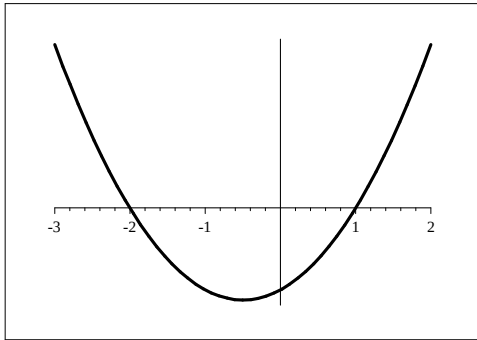
Example 2

Solve $x^2 + x + 1 > 3$ and write your answer in interval notation

Solution

$$x^2 + x + 1 > 3 \Rightarrow x^2 + x - 2 > 0 \Rightarrow (x-1)(x+2) > 0$$

Now sketch the graph of $y = (x-1)(x+2)$



Using the graph:

$$(x - 1)(x + 2) > 0 \Rightarrow x < -2 \text{ or } x > 1 \Rightarrow x \in (-\infty, -2) \cup (1, \infty)$$

Example 3

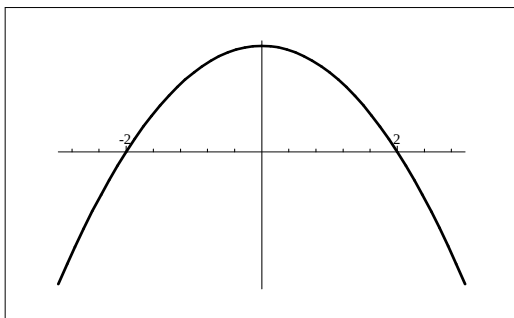
Solve $\frac{4 - x^2}{x^2} > 0$ and write your answer in interval notation

Solution

$$\frac{4 - x^2}{x^2} > 0 \Rightarrow 4 - x^2 > 0 \text{ and } x^2 \neq 0 \text{ (because } x^2 \geq 0 \text{ for all } x)$$

$$\Rightarrow (2 - x)(2 + x) > 0 \text{ and } x^2 \neq 0$$

Now sketch the graph of $y = (2 - x)(2 + x)$



Using the graph:

$$(2 - x)(2 + x) > 0 \text{ and } x^2 \neq 0$$

$$\Rightarrow x \in (-2, 2) \text{ and } x^2 \neq 0$$

$$\Rightarrow x \in (-2, 0) \cup (0, 2)$$